# Atomic clocks

PHYS4055 Atomic Physics and Quantum Optics

Week 13

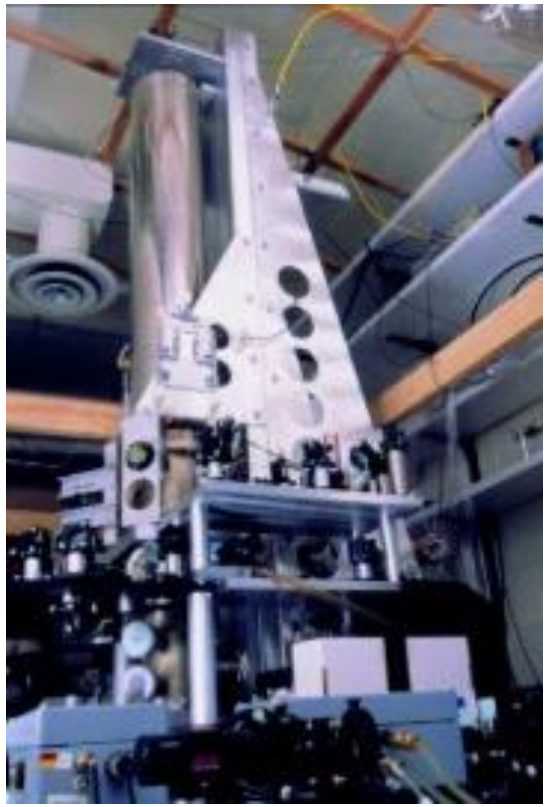


1/2 swing per second
(loses 1 minute a day)




50000 vibrations per second

(loses 1 second a month)



9192631770 periods per second
(NIST Cs clock)
(loses 1 second in 300 mln years)

-all atoms are identical
-atoms do not “wear out”

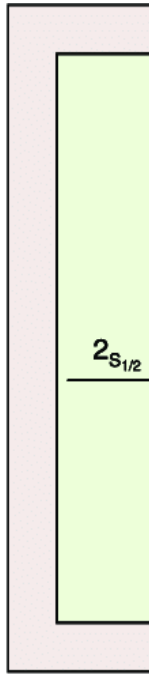
Clocks based on higher frequencies are more accurate because each oscillation (or "tick") represents a smaller unit of time, allowing for a finer-grained measurement of time and a reduced impact from counting errors.

# Cs atomic clock

# Cs atomic clock

$^{133}Cs$: [Xe]6s $^2S_{1/2}$

L=0, J=1/2
I=7/2



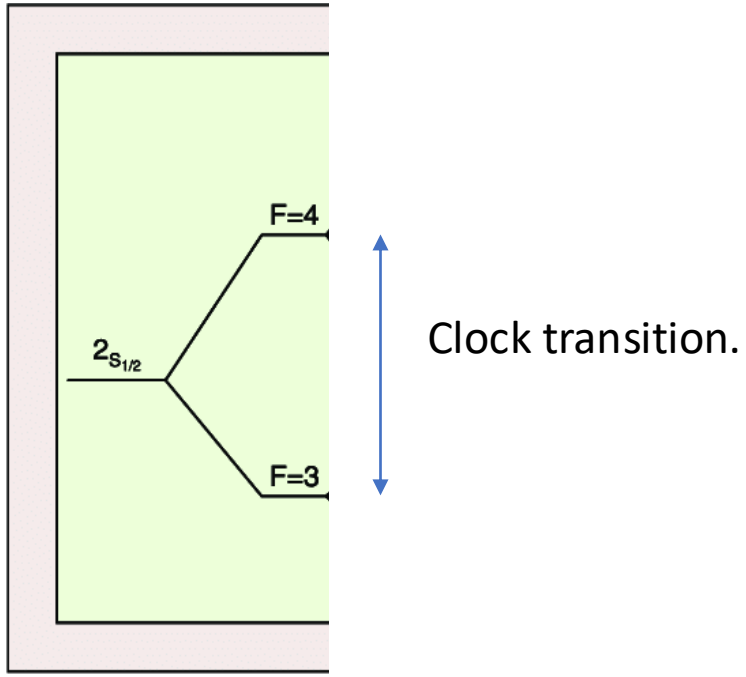
# Cs atomic clock

$^{133}$Cs: [Xe]6s $^{2}S_{1/2}$

L=0, J=1/2
I=7/2

I>J, hence will split into 2J+1=2 hyperfine levels:

F=|I+J|…|I-J|=3,4




Clock transition.

Goal: measure the frequency (9192631770 Hz, 3.26 cm, 0.307 $cm^{-1}$) of the F=3, $m_f$=0→ F=3, $m_f$=0 transition

# Zeeman effect on HFS

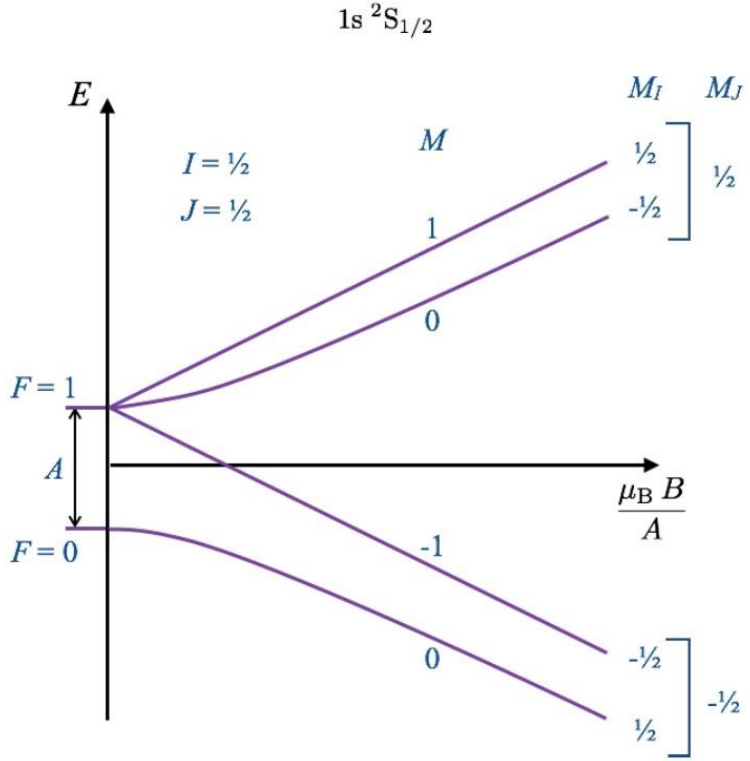
- Weak field:

$$E = g_F \mu_B B M_F$$

- Strong field:

$$E_{ZE} = g_J \mu_B B M_J + \langle I M_I J M_J | \, A \, \mathbf{I} \cdot \mathbf{J} \, | I M_I J M_J \rangle$$
$$= g_J \mu_B B \, M_J + A \, M_I \, M_J \, .$$

J=1/2, I=1/2
F=0,1

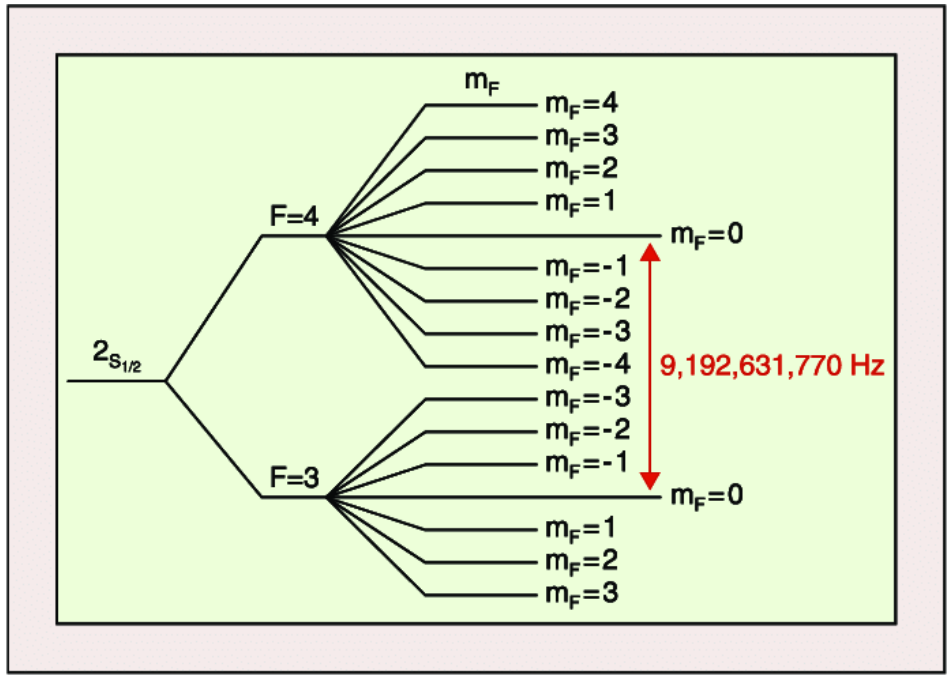


# Cs atomic clock

$^{133}$Cs: [Xe]6s $^2S_{1/2}$

L=0, J=1/2
I=7/2

Applying the magnetic field, we see Zeeman effect.



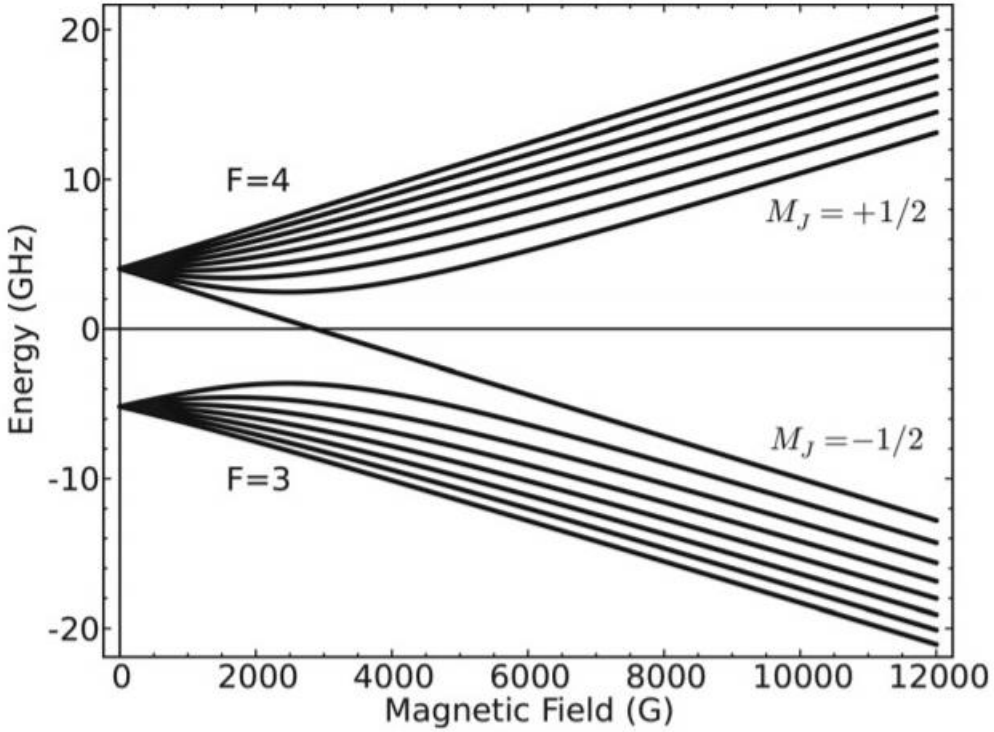
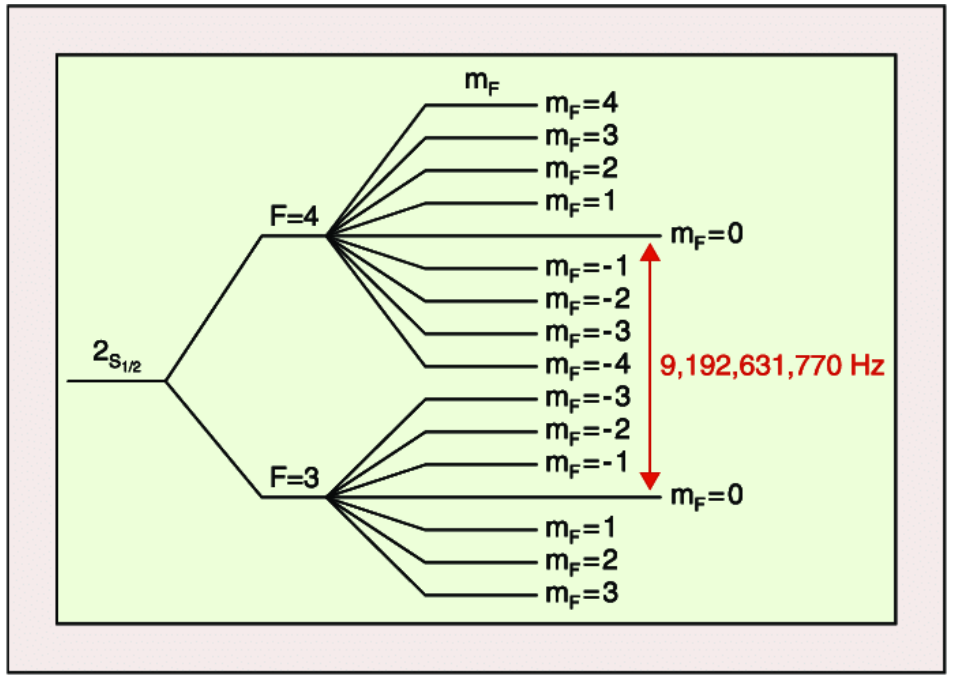




Fig. 2. Zeeman energy level pattern for Cs in its $6^2S_{1/2}$ ground electronic state.

# Cs atomic clock

Goal: measure the frequency of the F=3, $m_f$=0→ F=3, $m_f$=0 transition



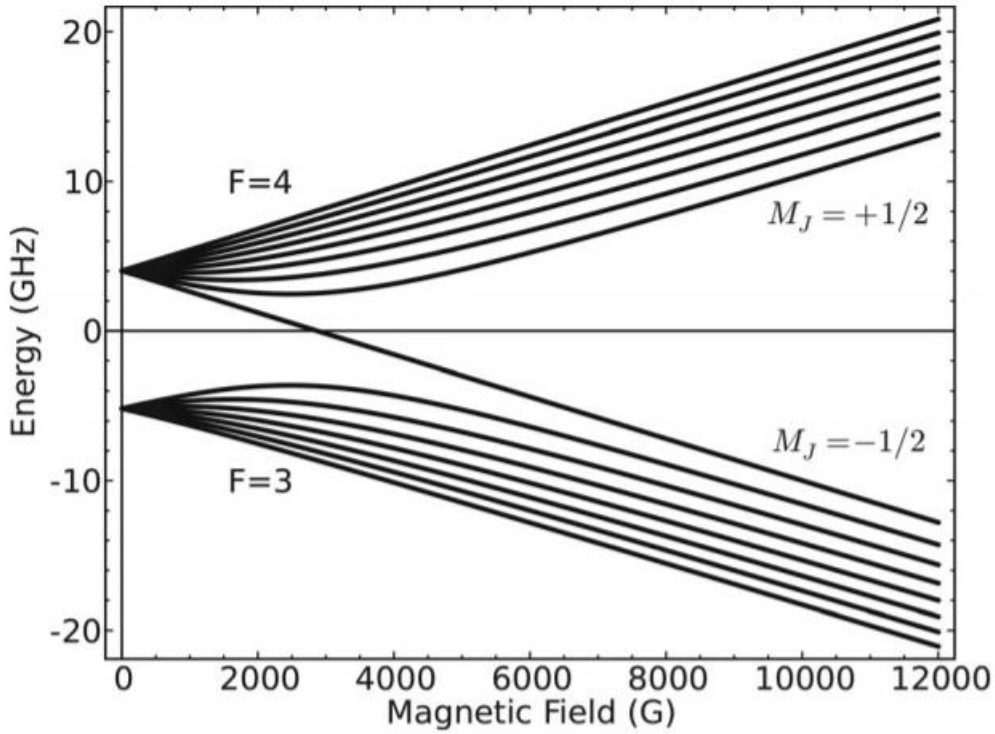
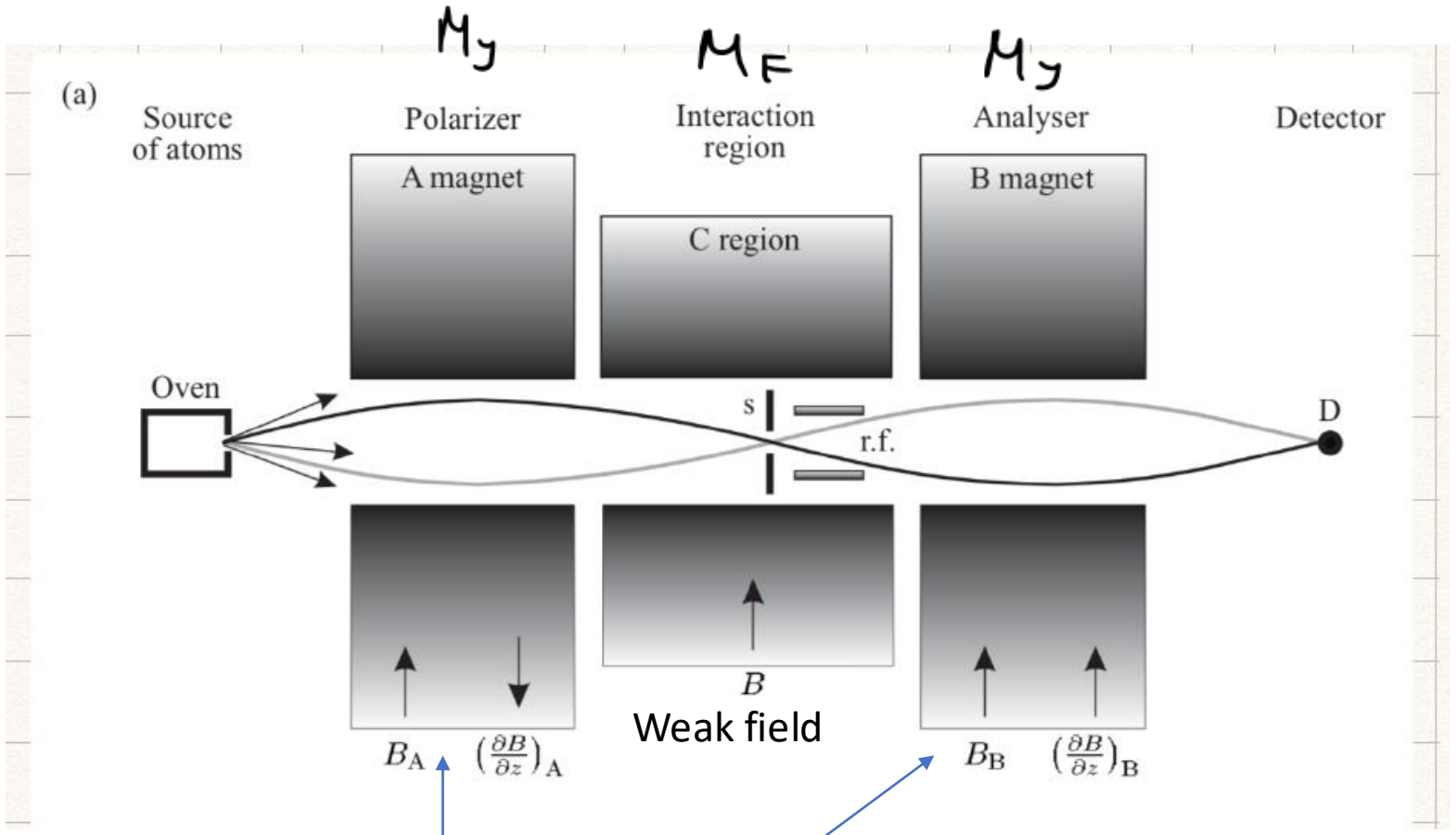



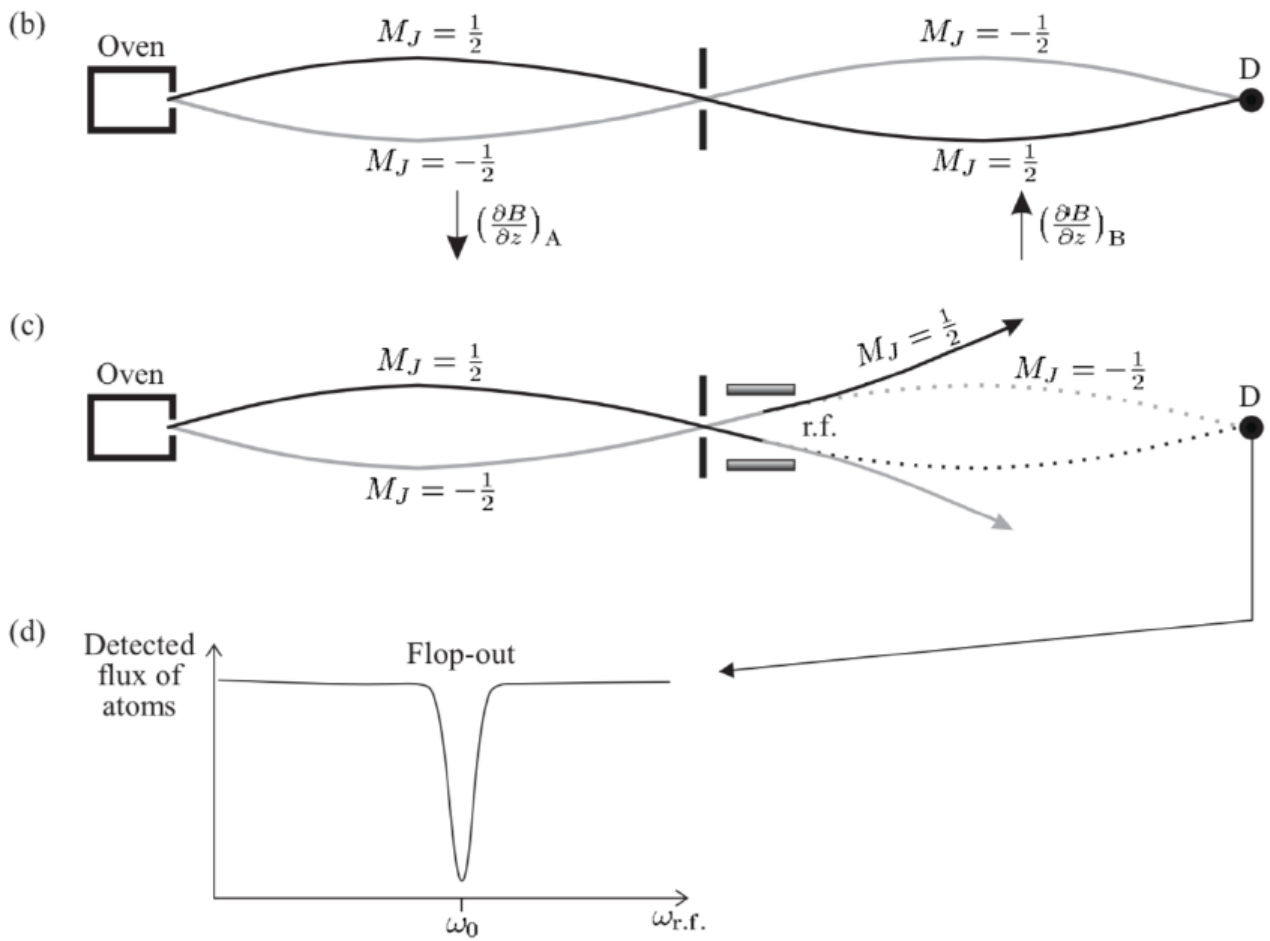

Fig. 2. Zeeman energy level pattern for Cs in its $6^2S_{1/2}$ ground electronic state.

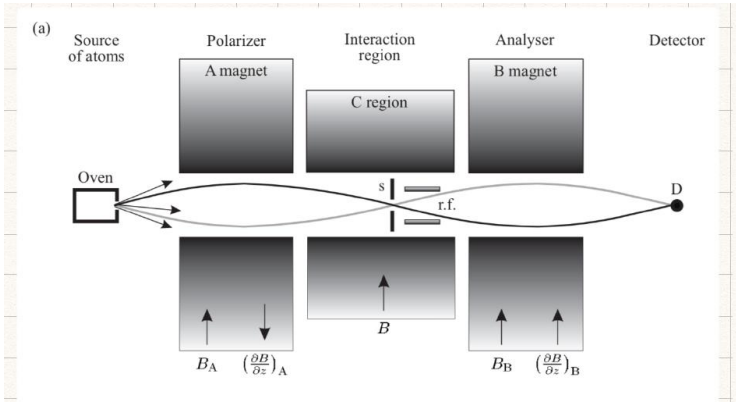
# Atomic beam technique



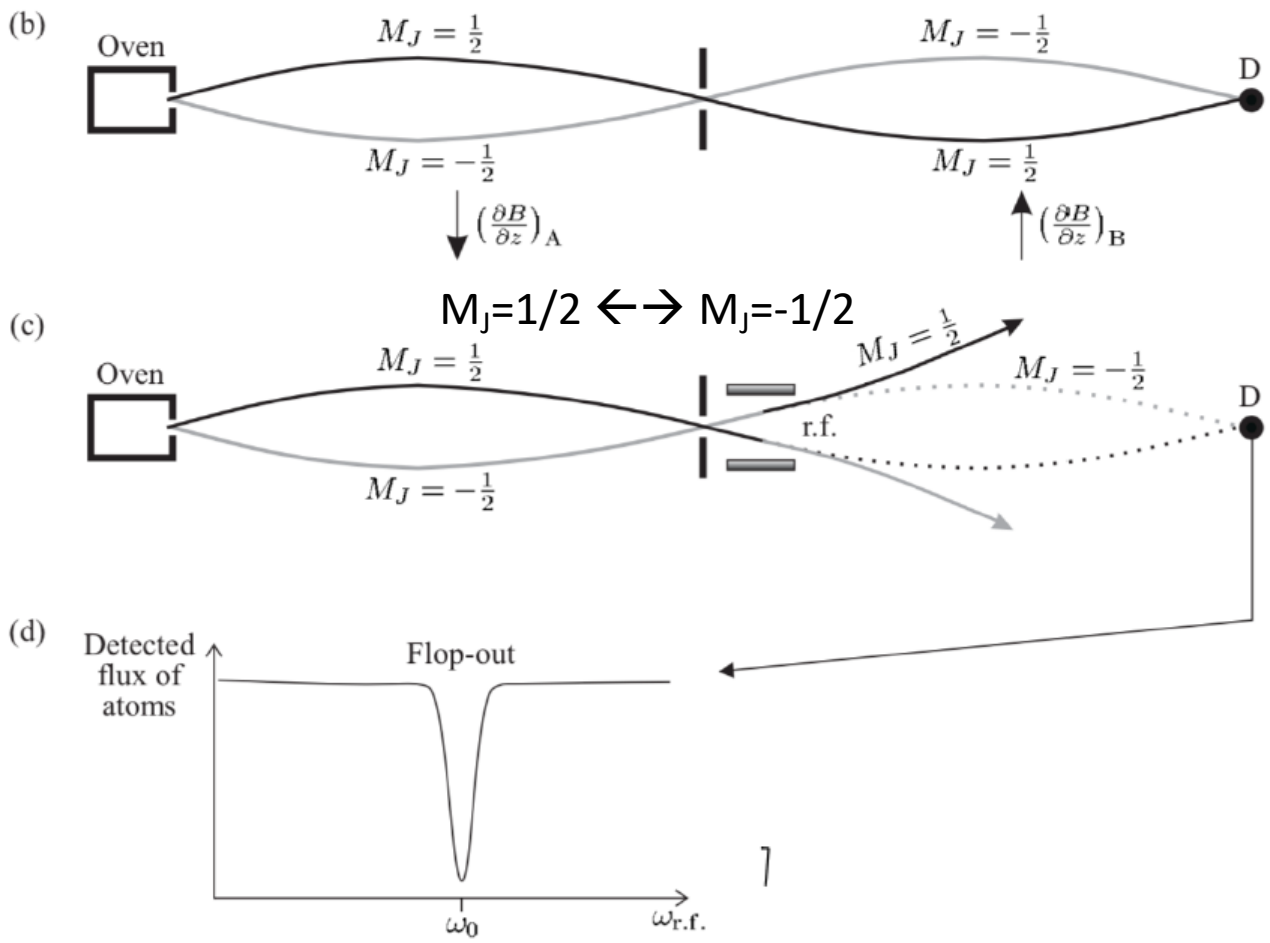
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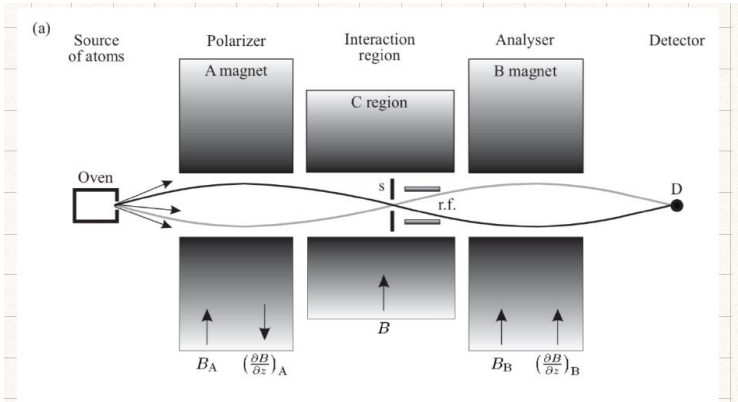
$$\text{Force} = -\frac{\mathrm{d}E_{\mathrm{ZE}}}{\mathrm{d}z} = g_J \mu_{\mathrm{B}} \frac{\mathrm{d}B}{\mathrm{d}z} M_J$$



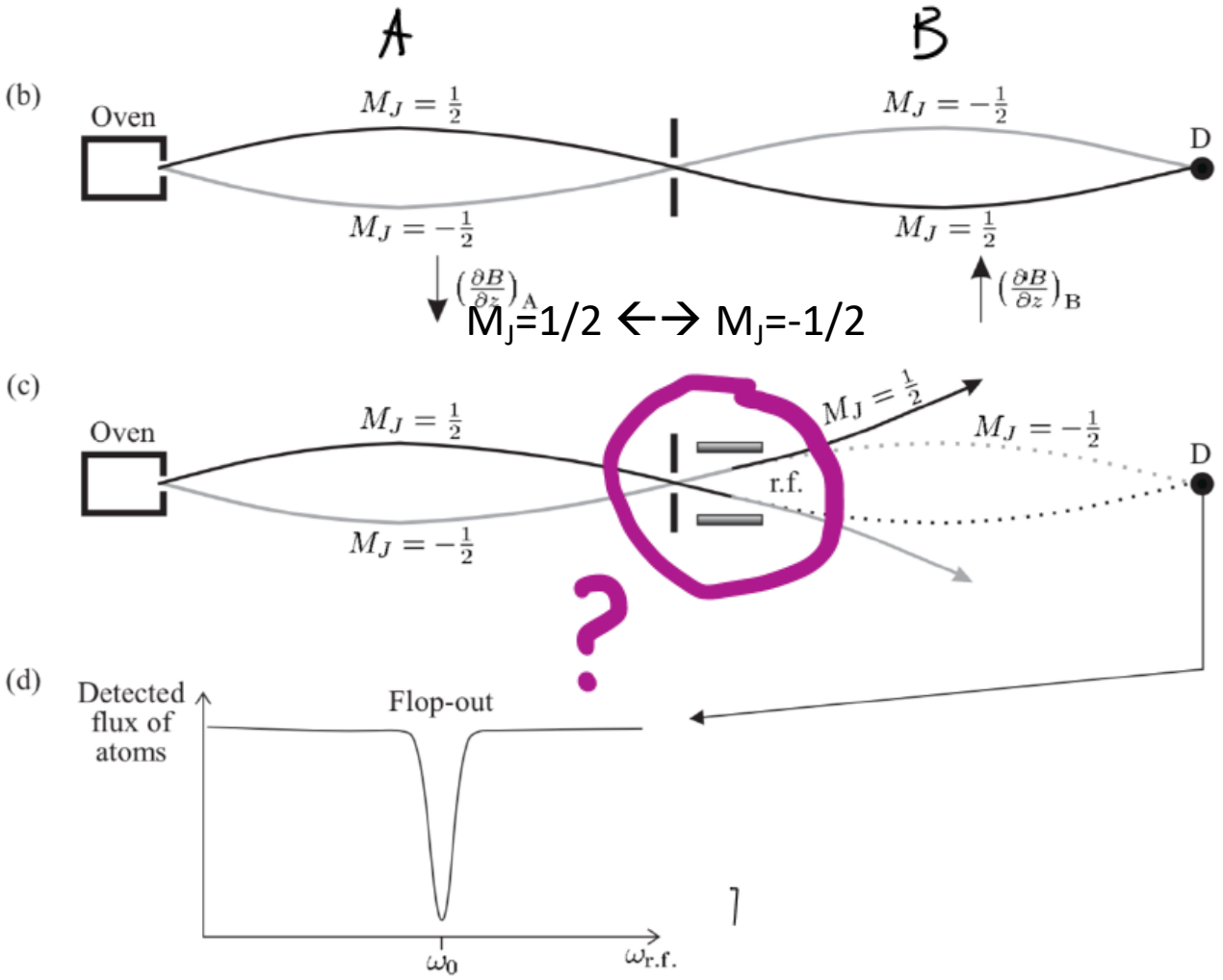



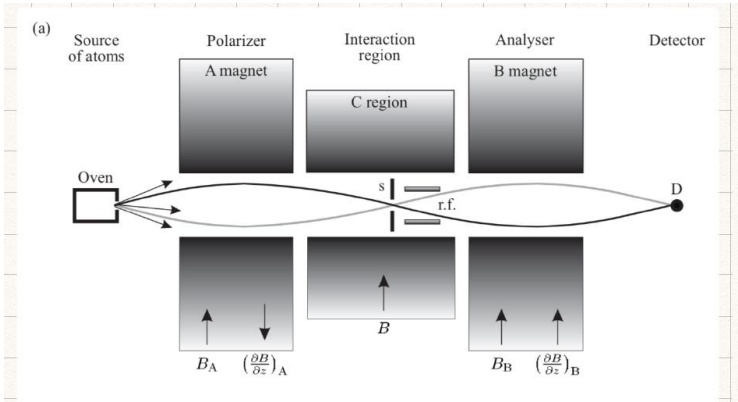
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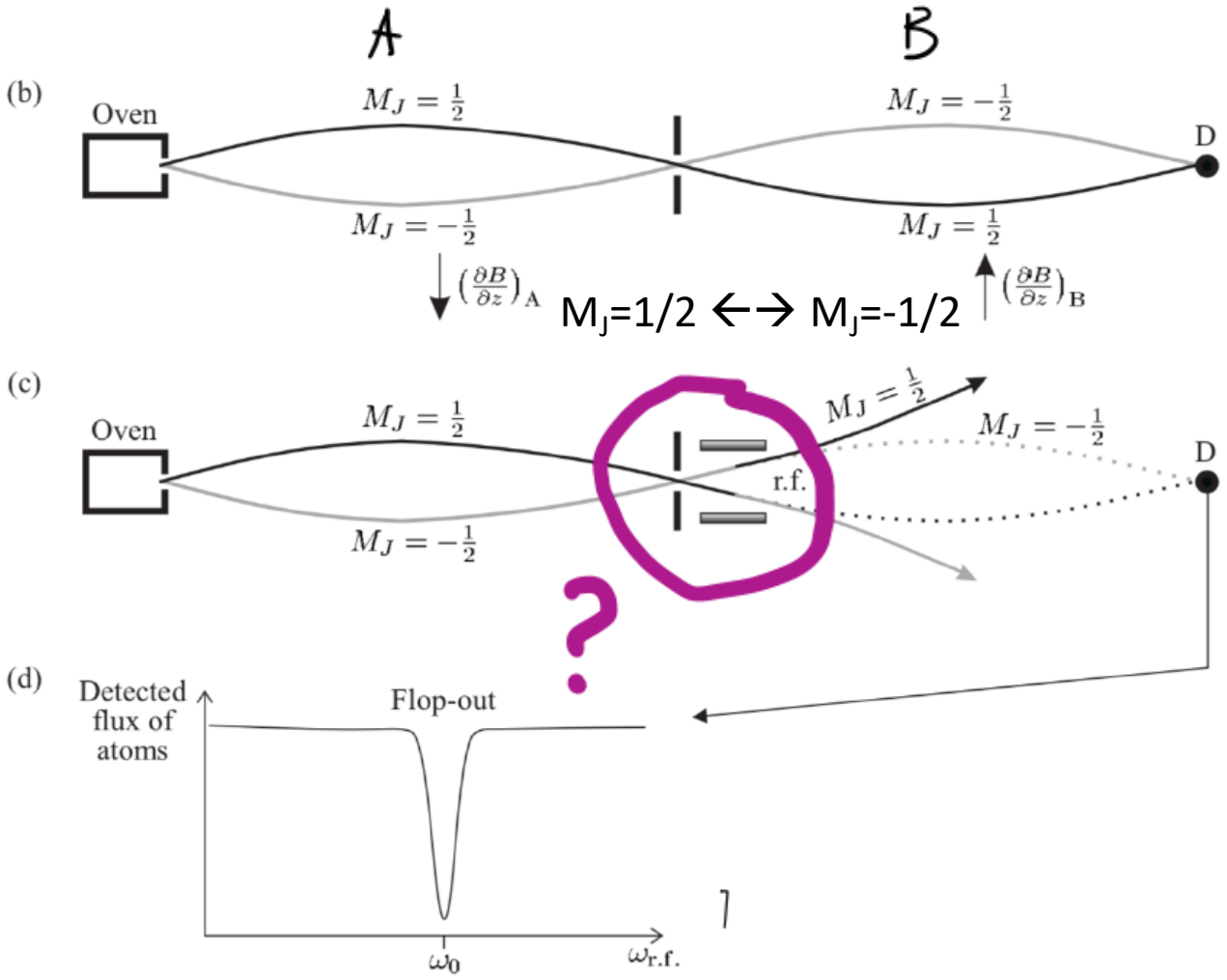



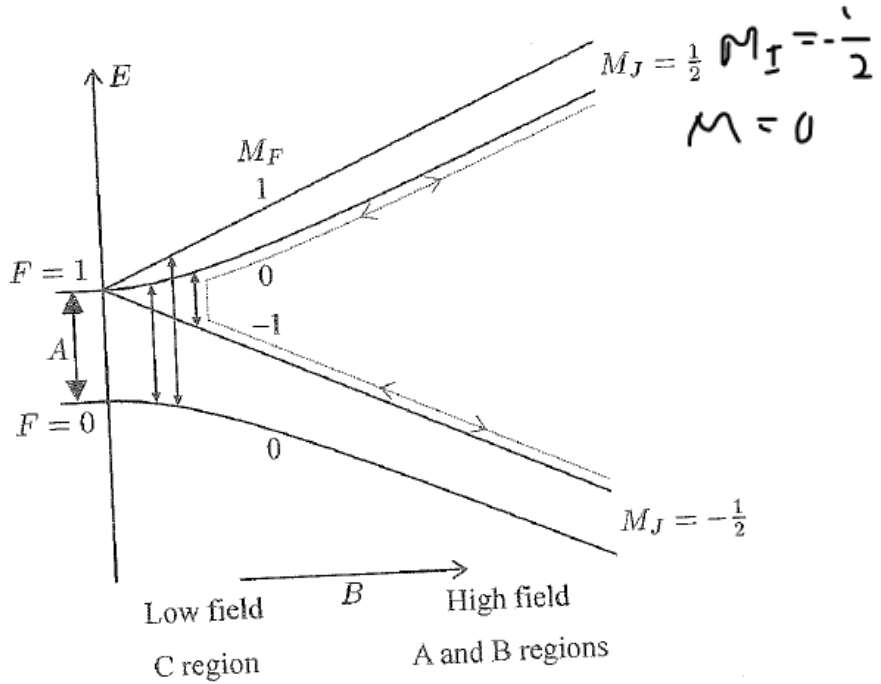
# Atomic beam technique



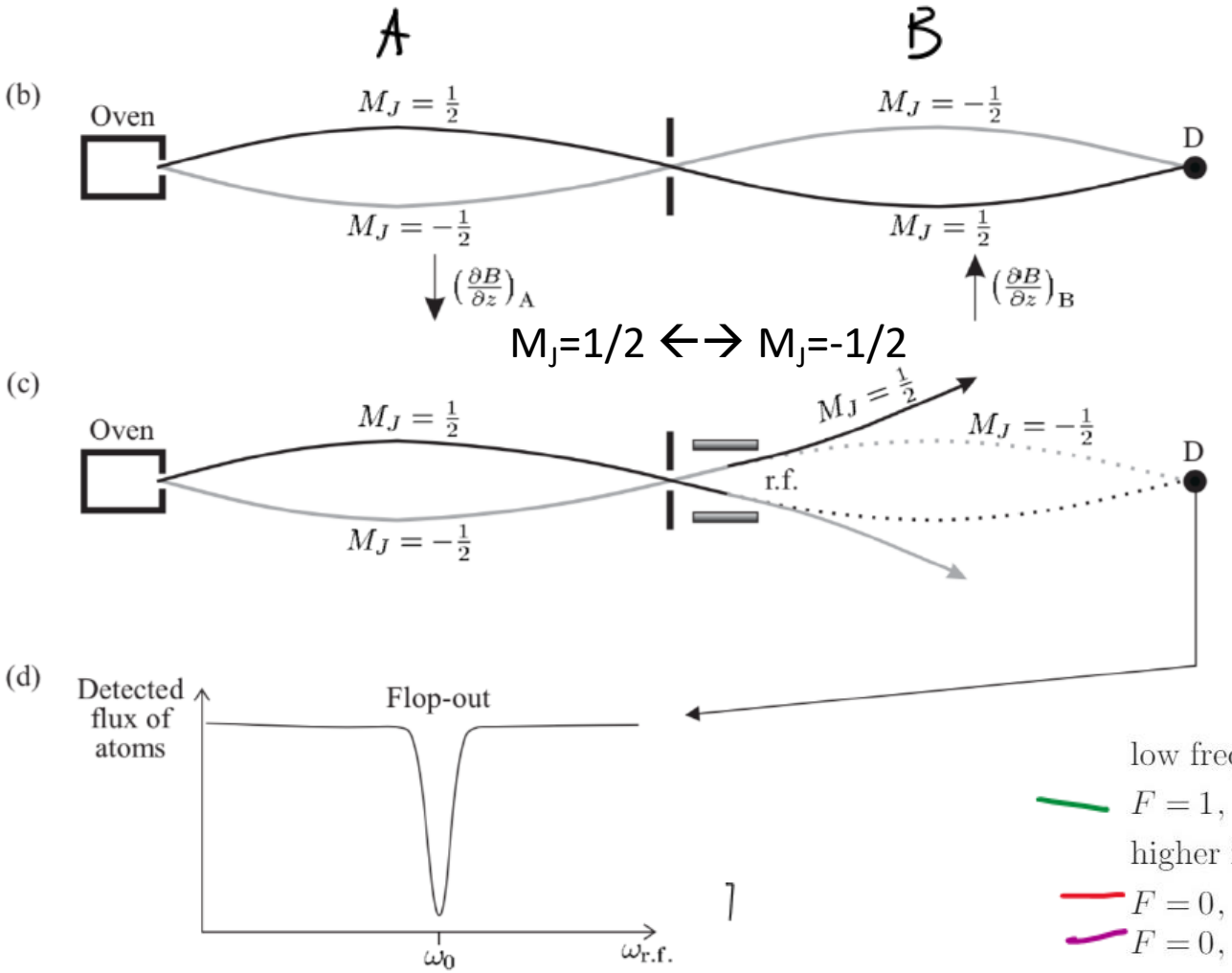



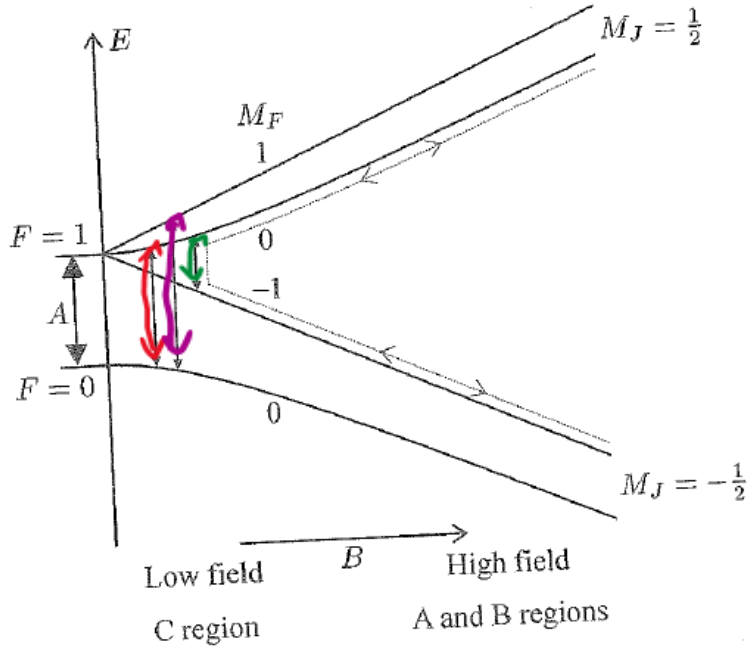
# Atomic beam technique



# Atomic beam technique




low frequency ($\Delta F = 0$):

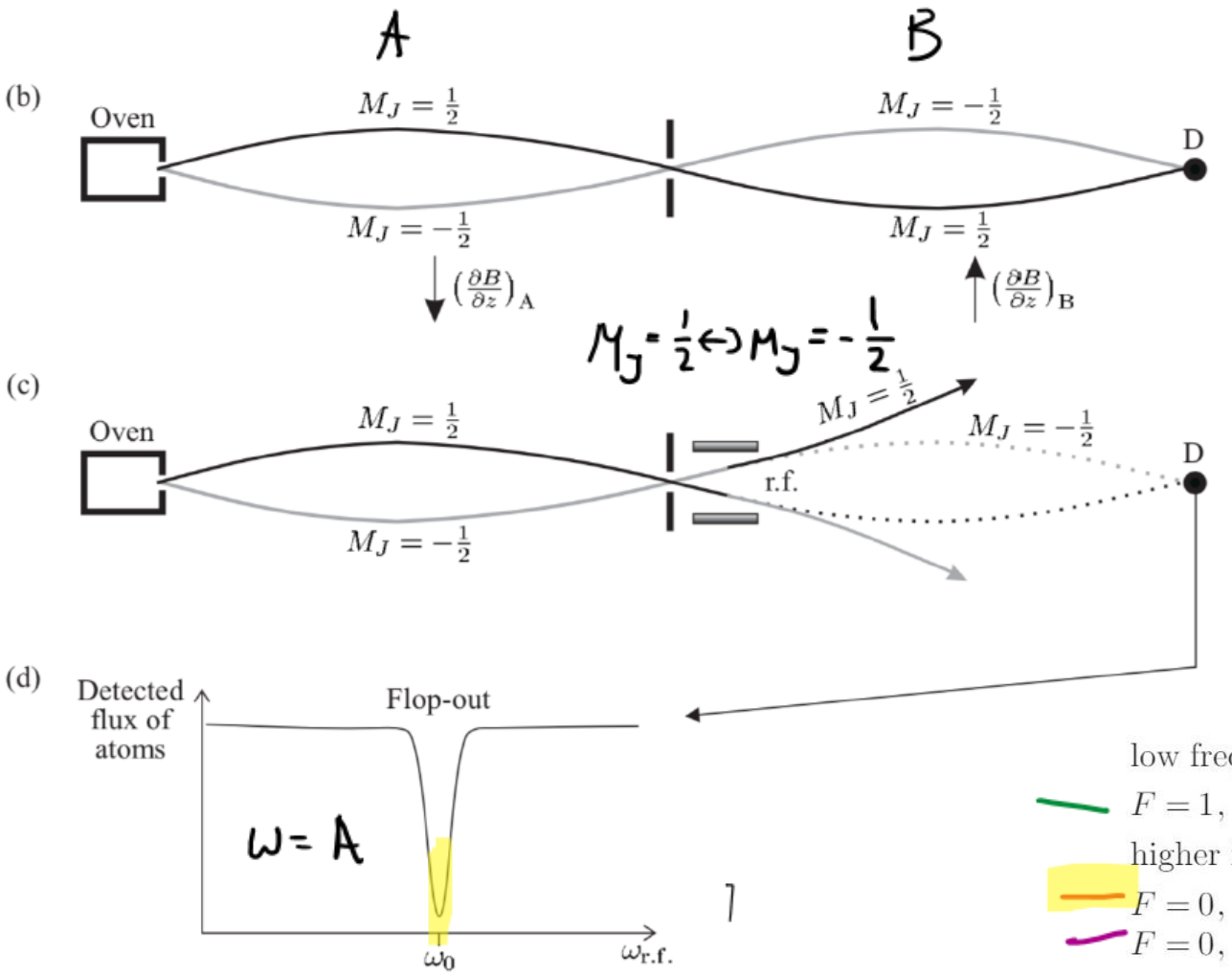
$F = 1, \quad M_F = 0 \quad \leftrightarrow \quad F = 1, \quad M_F = -1 \quad \text{with} \quad \Delta E = g_F \mu_B B\,;$

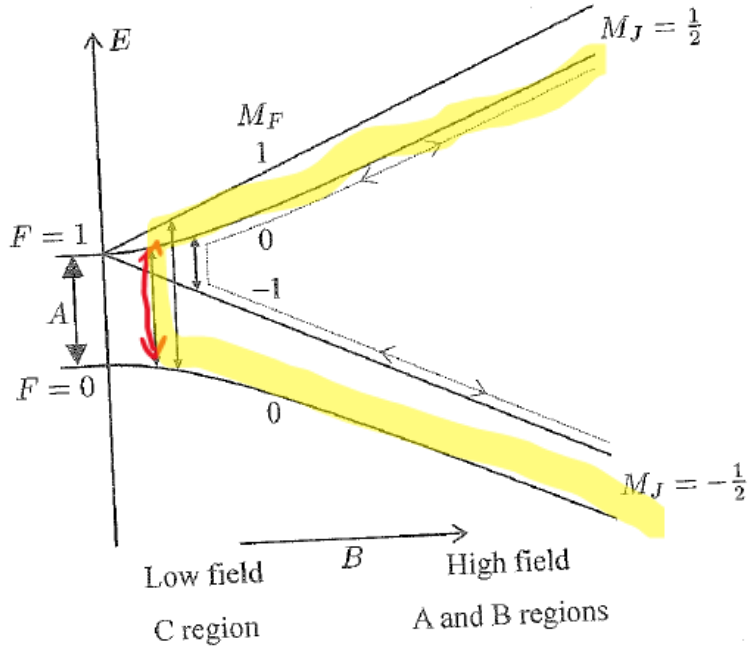
higher frequency ($\Delta F = \pm 1$):

$F = 0, \quad M_F = 0 \quad \leftrightarrow \quad F = 1, \quad M_F = 0 \quad \text{with} \quad \Delta E = A\,,$

$F = 0, \quad M_F = 0 \quad \leftrightarrow \quad F = 1, \quad M_F = 1 \quad \text{with} \quad \Delta E = A + g_F \mu_B B\,.$

# Atomic beam technique




low frequency ($\Delta F = 0$):

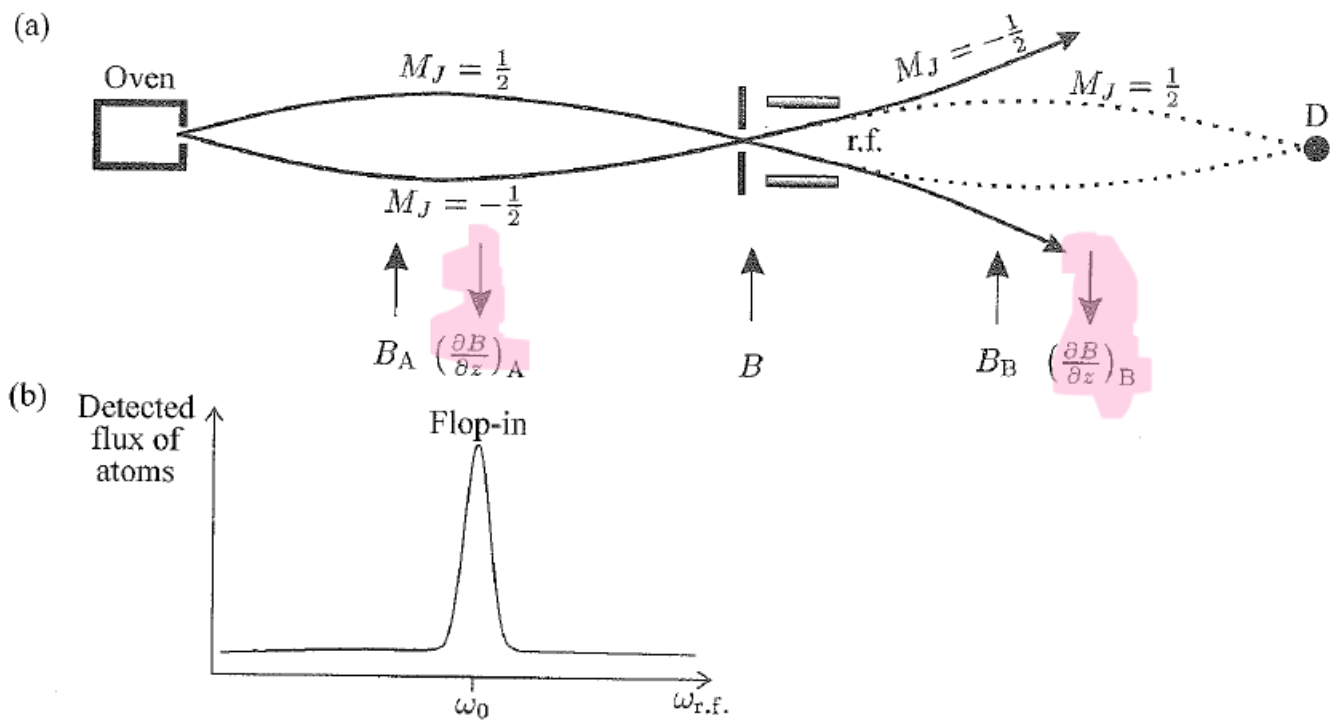
$F = 1, \quad M_F = 0 \quad \leftrightarrow \quad F = 1, \quad M_F = -1 \quad \text{with} \quad \Delta E = g_F \mu_B B$;

higher frequency ($\Delta F = \pm 1$):

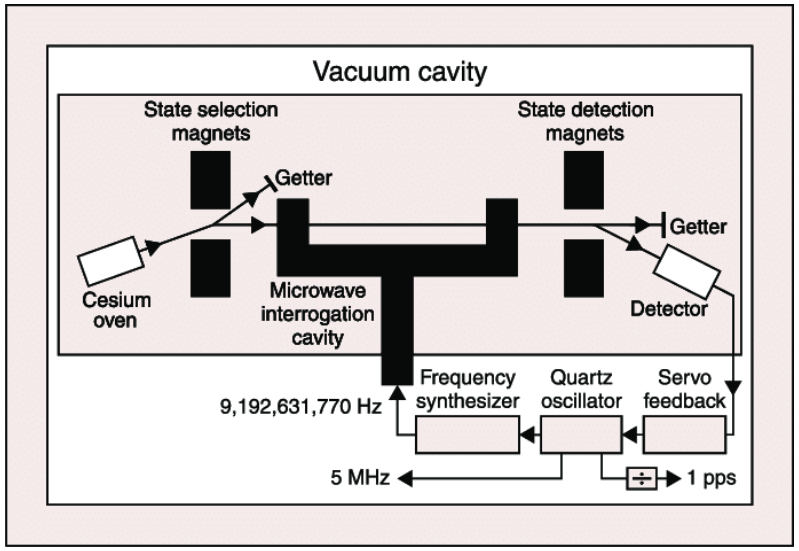
$F = 0, \quad M_F = 0 \quad \leftrightarrow \quad F = 1, \quad M_F = 0 \quad \text{with} \quad \Delta E = A$,

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# Atomic beam technique



# Cs atomic clock



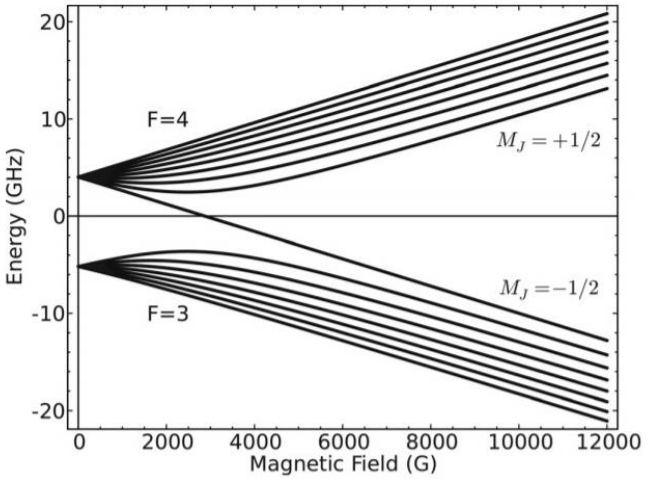
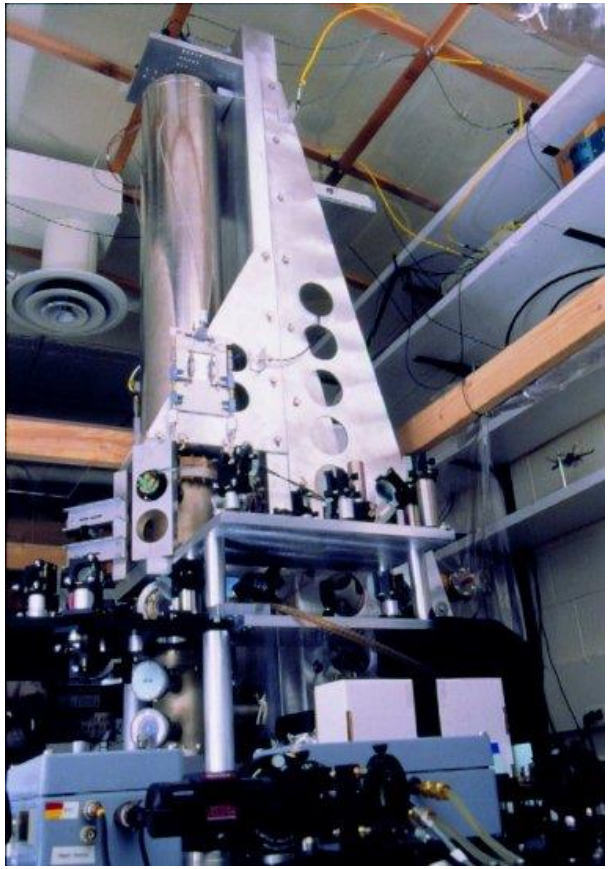
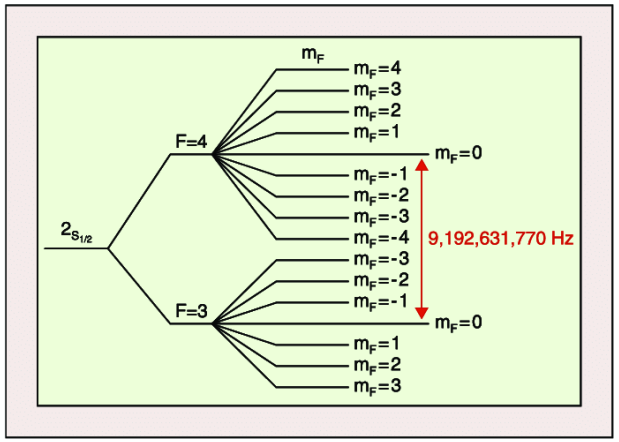




Fig. 2. Zeeman energy level pattern for Cs in its $6^2S_{1/2}$ ground electronic state.

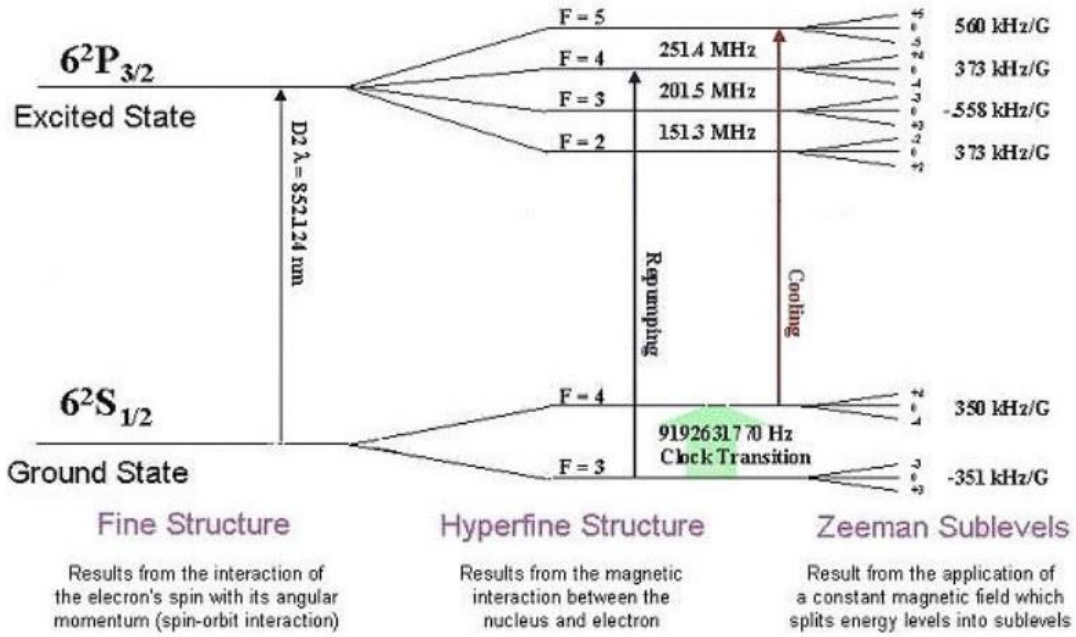




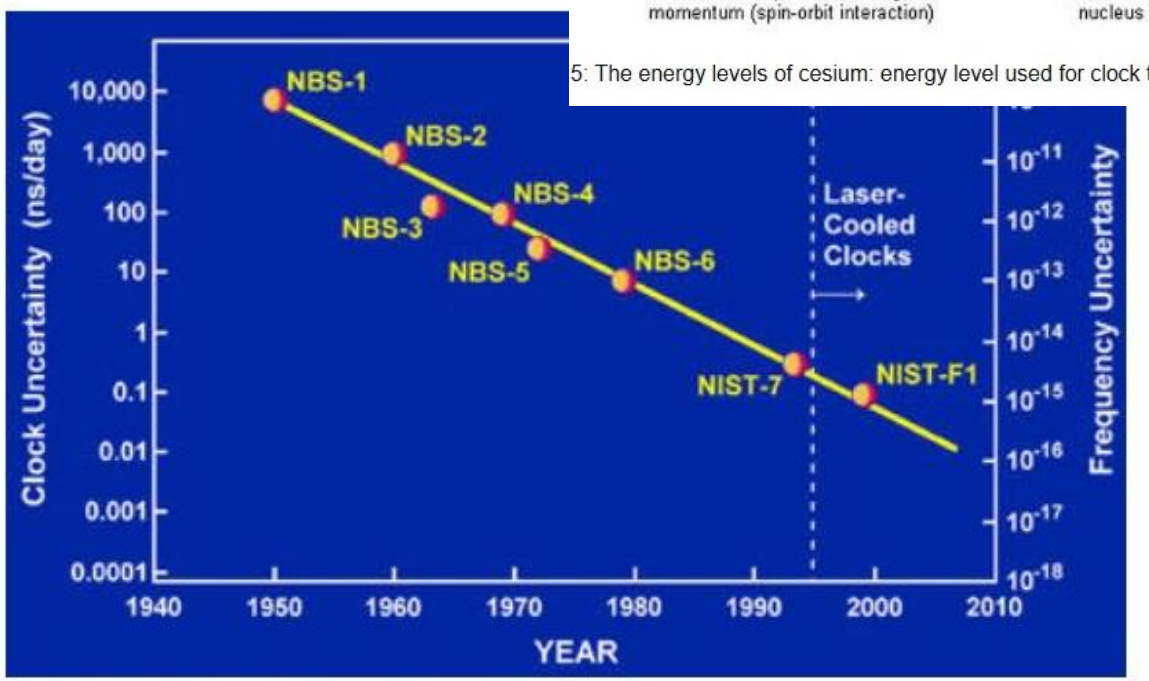

https://www.youtube.com/watch?v=MGoVXLzUDsQ&ab_channel=NationalPhysicalLaboratory

# The second is defined based on the Cs clock

- **Original definition:** 1⁄86,400 of a mean solar day
- **1956:** The second was redefined in terms of a year. The second was thus defined as "the fraction 1⁄31,556,925.9747 of the tropical year for 1900 January 0 at 12 hours ephemeris time"
- **1967:** The second is the duration of 9192631770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the caesium-133 atom.
- **2019 definition:** The second, symbol s, is the SI unit of time. It is defined by taking the fixed numerical value of the caesium frequency $\Delta\nu_{Cs}$, the unperturbed ground-state hyperfine transition frequency of the caesium-133 atom, to be 9192631770 when expressed in the unit Hz, which is equal to $s^{-1}$.
- https://www.youtube.com/watch?v=9ikbD7UGzoI&ab_channel=NationalInstituteofStandardsandTechnology




5: The energy levels of cesium: energy level used for clock transition and cooling.




Clock uncertainty improvements over the years

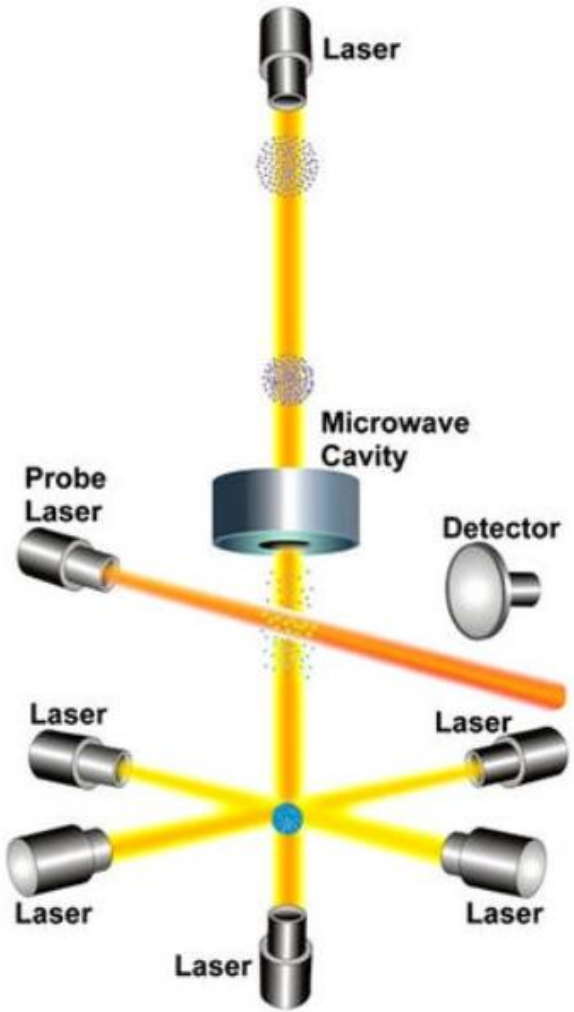



Diagram of a fountain atomic clock

# Optical clocks

# Sr atomic clock

Sr [Kr]$5s^2$, $^1S_0$ ground state.
Spin-forbidden clock transition, long lifetime, allows for precise frequency measurement

$$\Delta E \, \Delta t \leq \frac{\hbar}{2}$$

$$\Gamma = 6 \cdot 10^{-3} \frac{1}{s}$$

Double cooling: first "blue MOT", 461 nm cooling from 800 K to mK
Next the "red MOT", 689 nm (forbidden transition), down to μK.

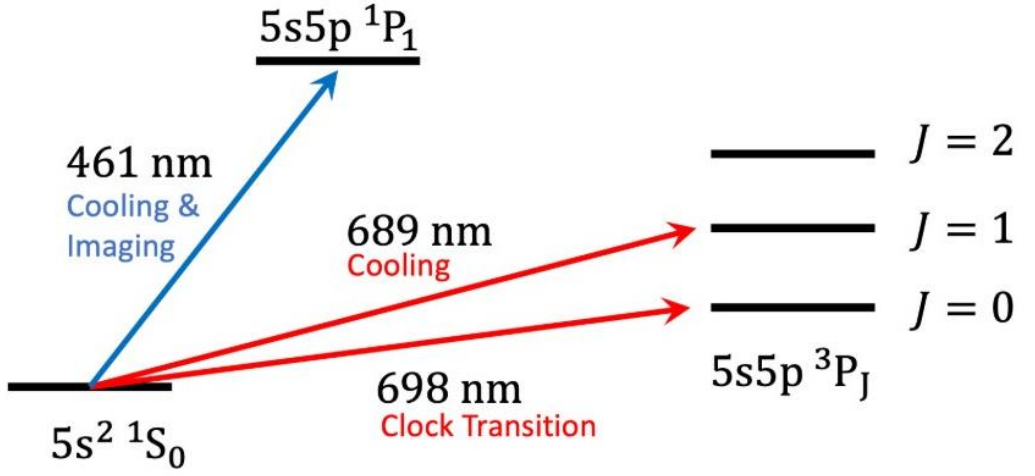
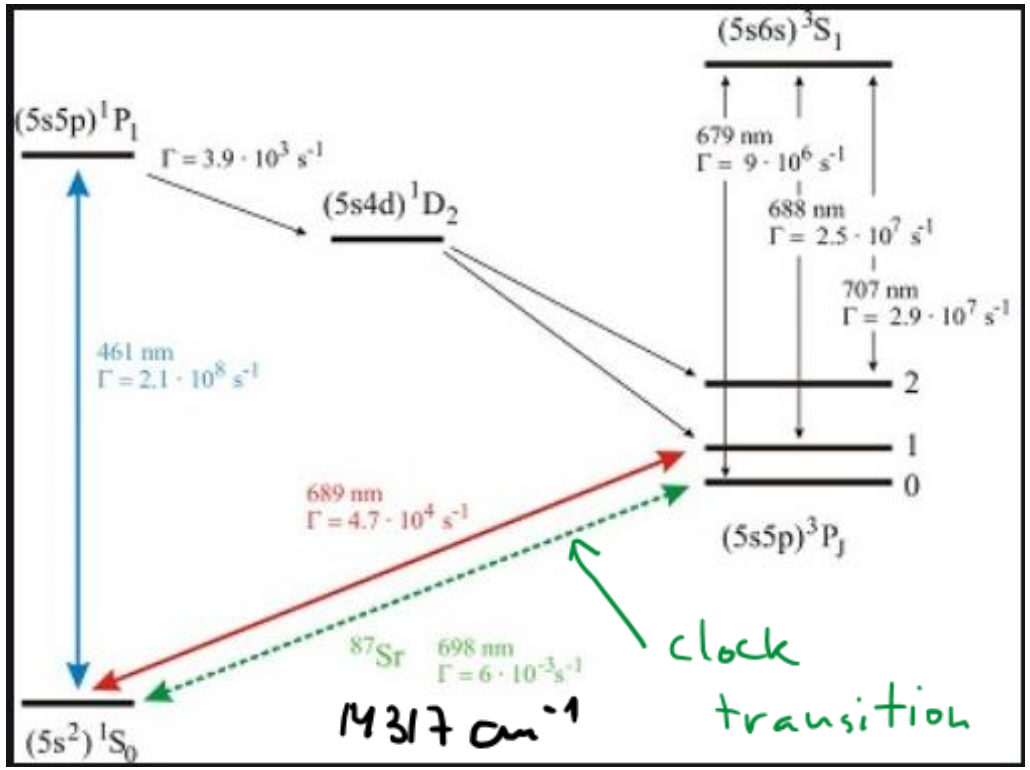



Fig. 1: the energy-level structure of Strontium.

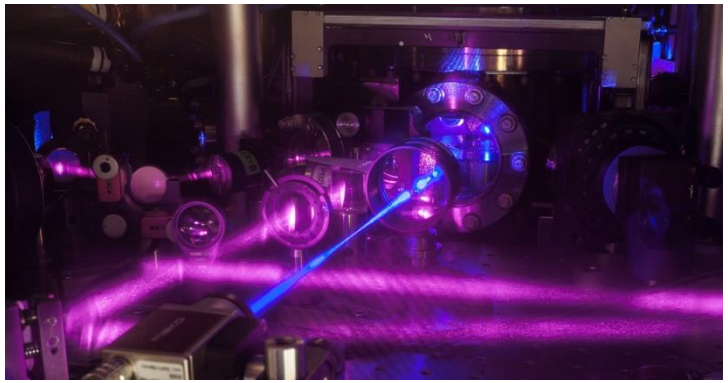


- Higher frequency compared to caesium time standard (429228 GHz vs 9.2 GHz)

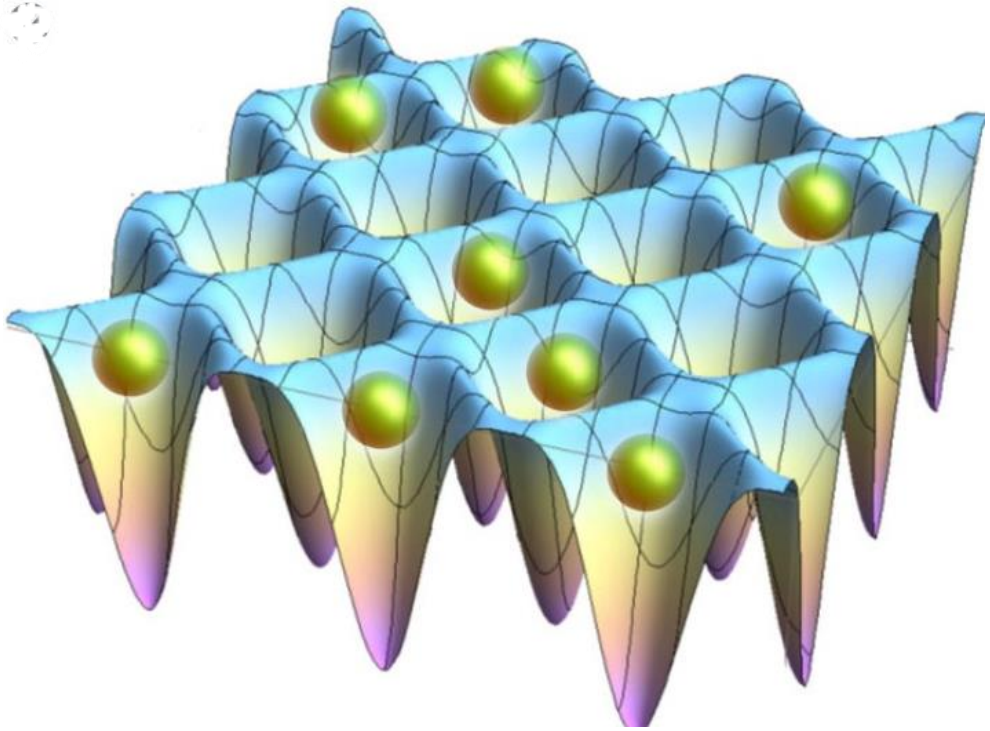
$$\sigma = \frac{1}{2\pi f_0 \sqrt{N T_m \tau}}$$

$\sigma$- Allan variance (measure of frequency stability)
$f_0$ – transition frequency
N – number of atoms
$T_m$ – measurement time
$\tau$ – lifetime of the level

- Higher frequency compared to cesium time standard (429228 GHz vs p.2 GHz)

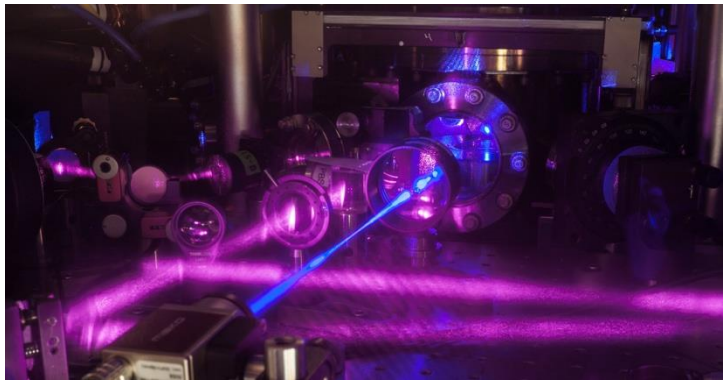


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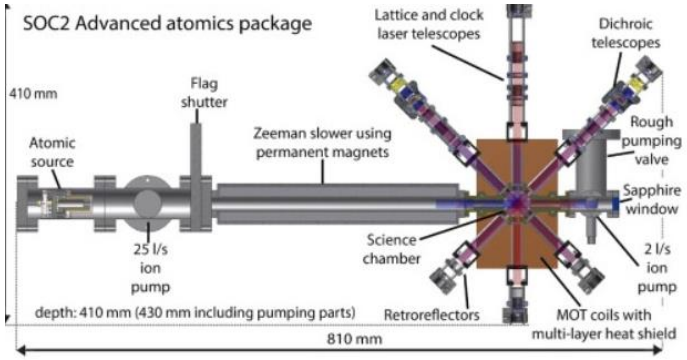


- Sr clock:
  - 10^4 atoms (large N)
  - Trapped in a lattice (large $T_m$)
  - Optical transition (large $f_0$)
  - Forbidden transition (long $\tau$)

- Higher frequency compared to cesium time standard (429228 GHz vs p.2 GHz)

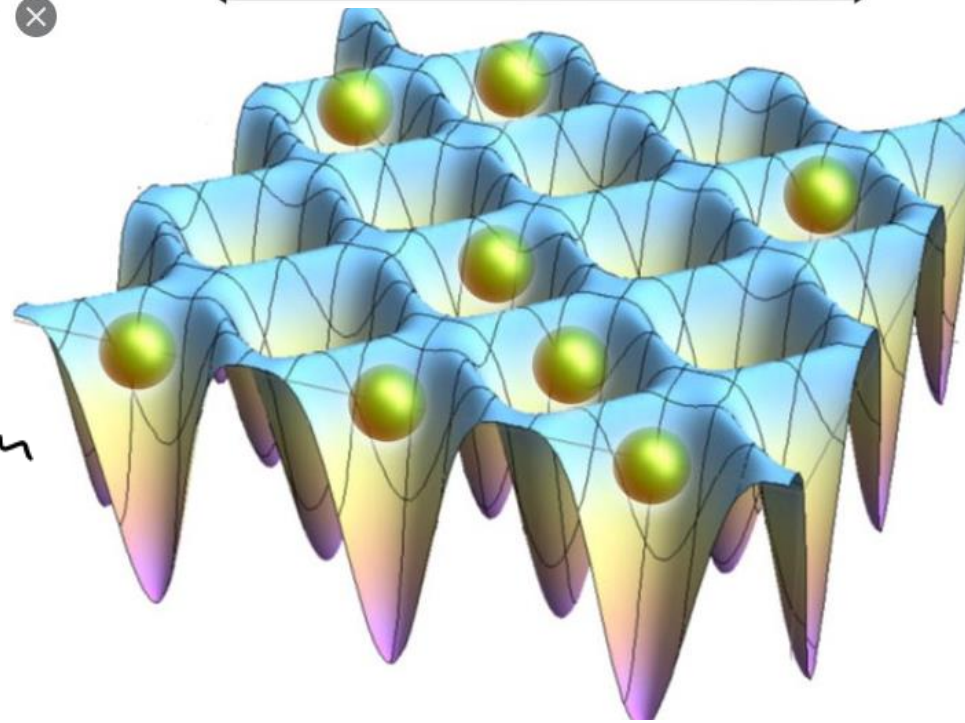


$$\sigma = \frac{1}{2\pi f_0 \sqrt{N T_m \tau}}$$



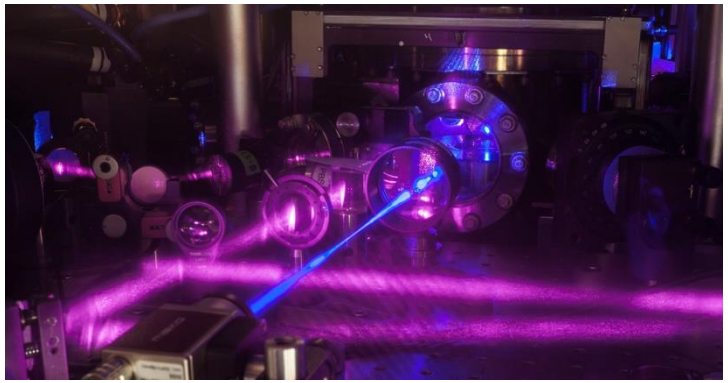

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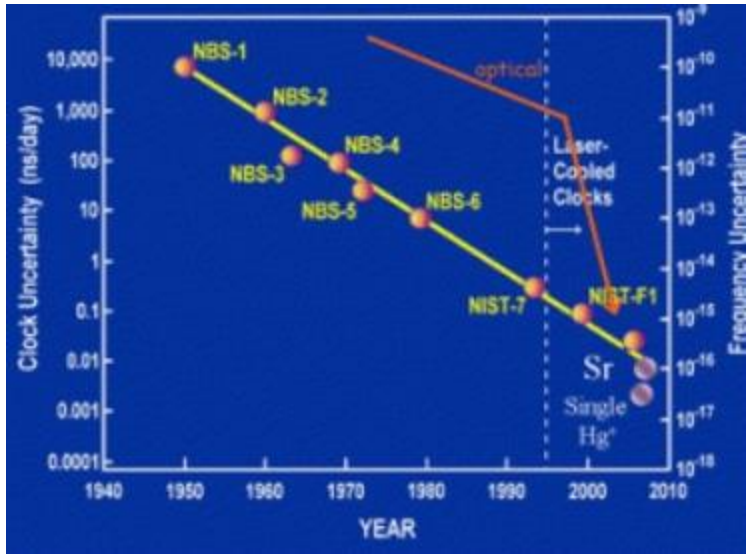
$5 \times 10^{-19}$ precision

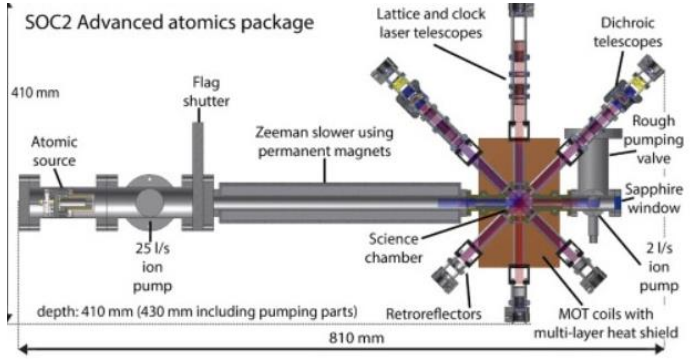


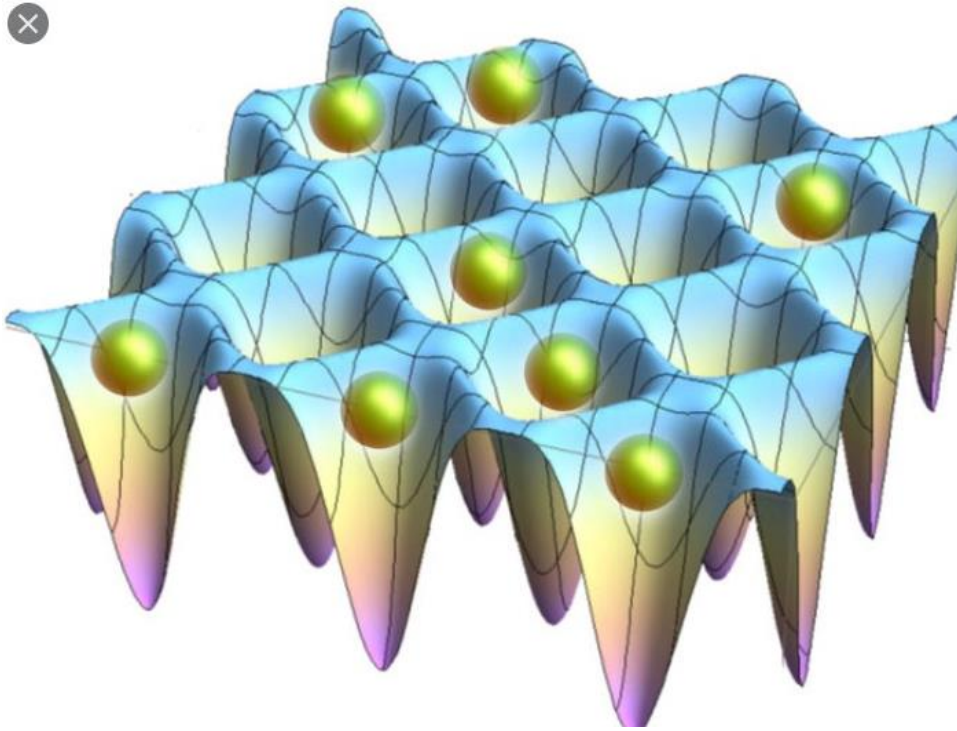
- https://www.youtube.com/watch?v=1JBHG6fipr4&ab_channel=Seeker

Strontium atomic clocks are considered to be one of the most precise and have been experimentally demonstrated to have a fractional frequency uncertainty in the order of ~ $10^{-19}$ . To put this in perspective, the clock must run longer than t of the universe for there to be an error of ~1 second in timekeeping!





- https://www.youtube.com/watch?v=1JBHG6fipr4&ab_channel=Seeker

Strontium atomic clocks are considered to be one of the most precise and have

NIST

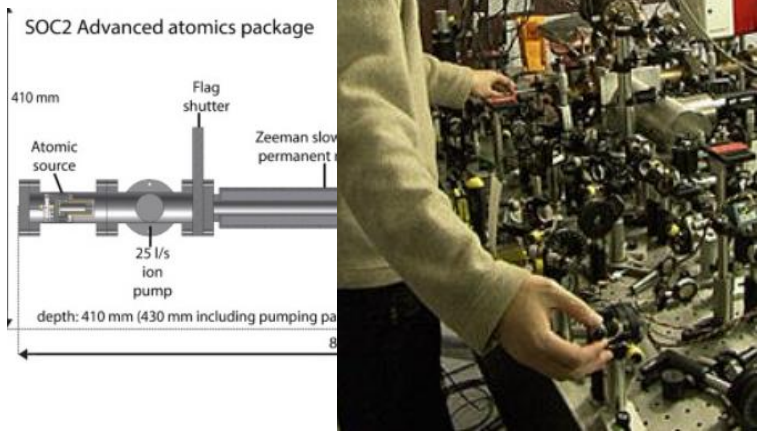
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NEWS

# NIST Ion Clock Sets New Record for Most Accurate Clock in the World

July 14, 2025






$Al^+$, quantum logic

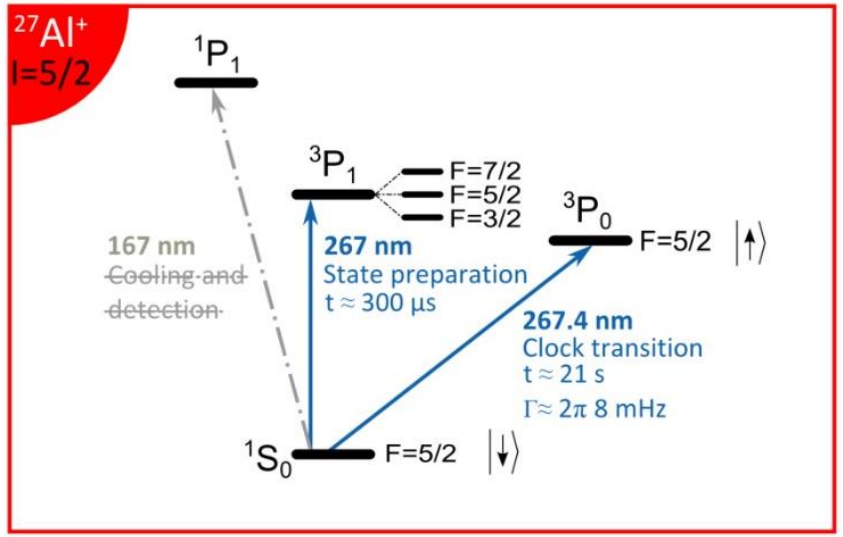
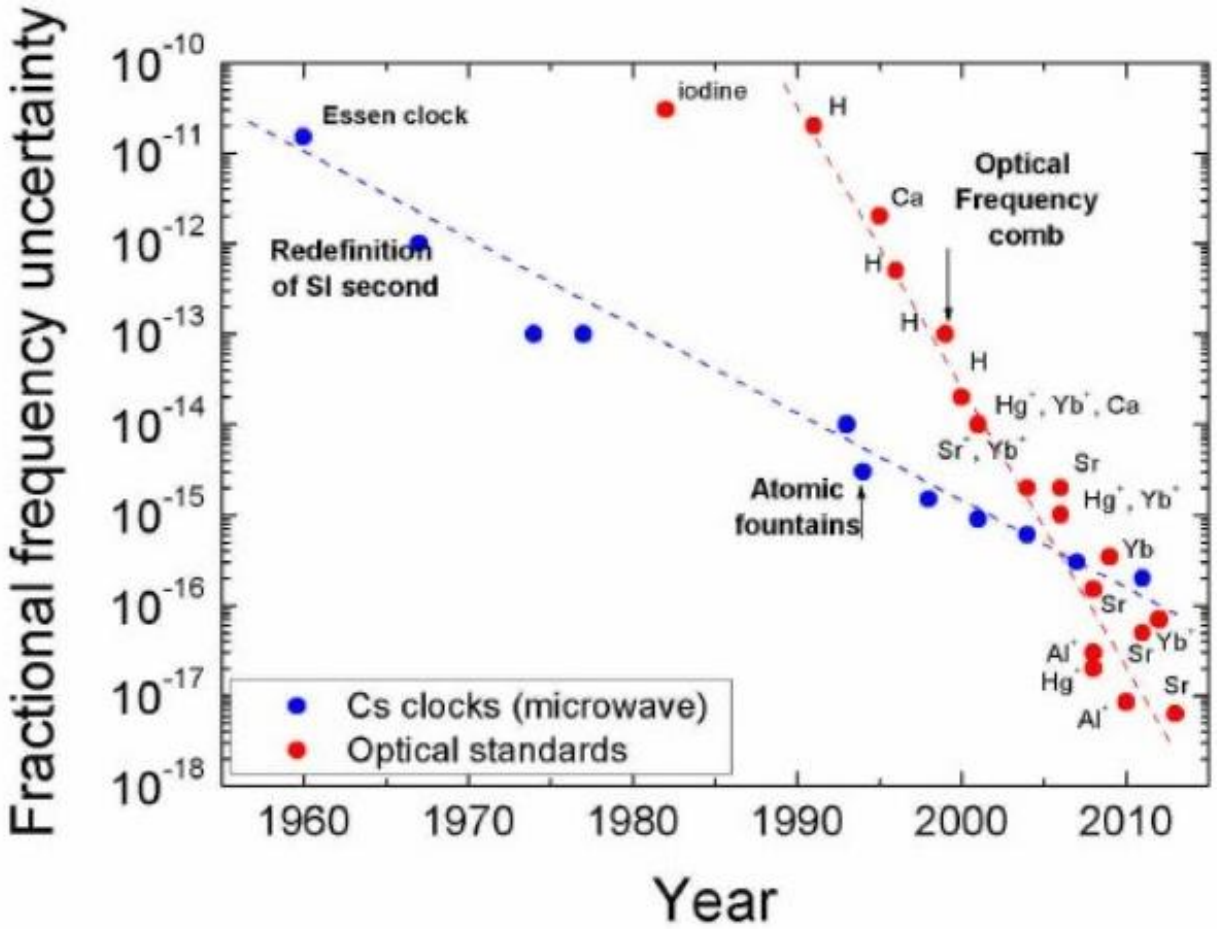
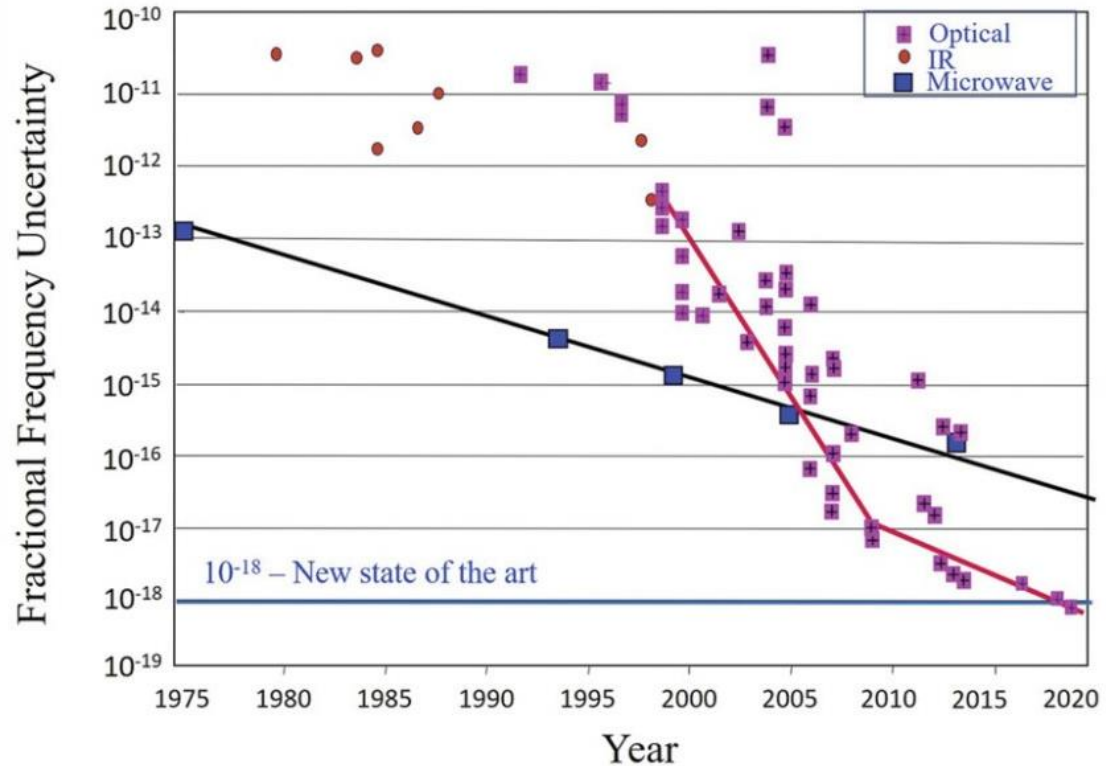



Figure 1: Relevant electronic levels and transitions in the $^{27}Al^+$ clock ion.




Historical accuracy of atomic clocks. Since 2000, the uncertainty of optical atomic clocks have rapidly improved to, and beyond, 1 part in $10^{18}$.

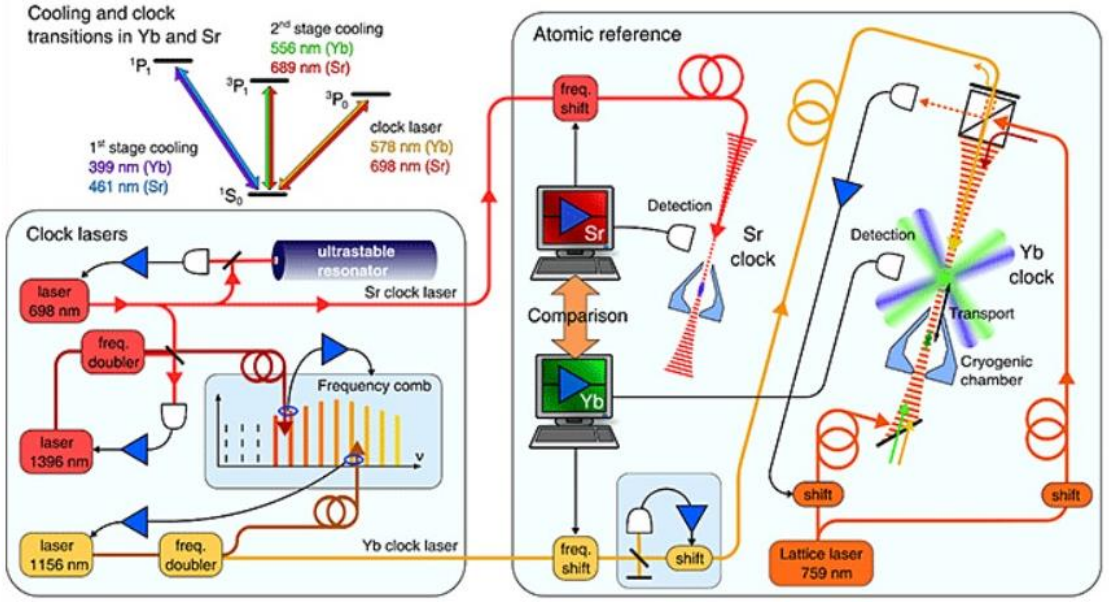


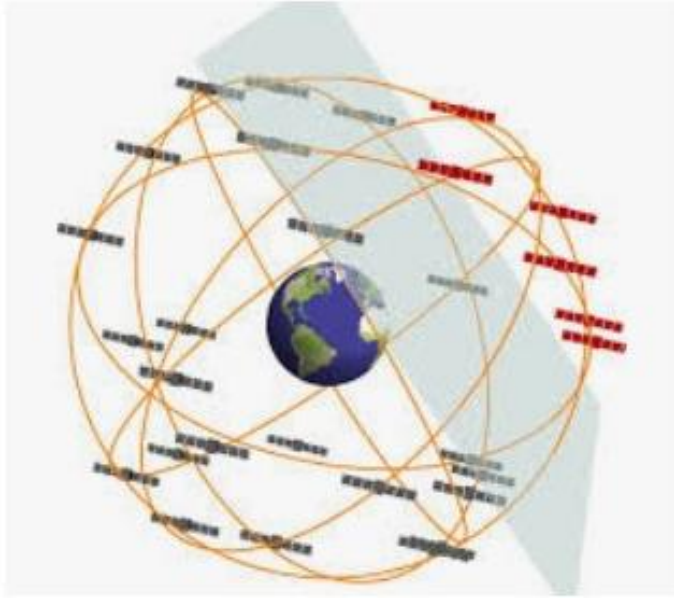
- Applications:
  - GPS
  - Geodesy
  - Quantum simulations
  - Time standards
  - Earthquake detection



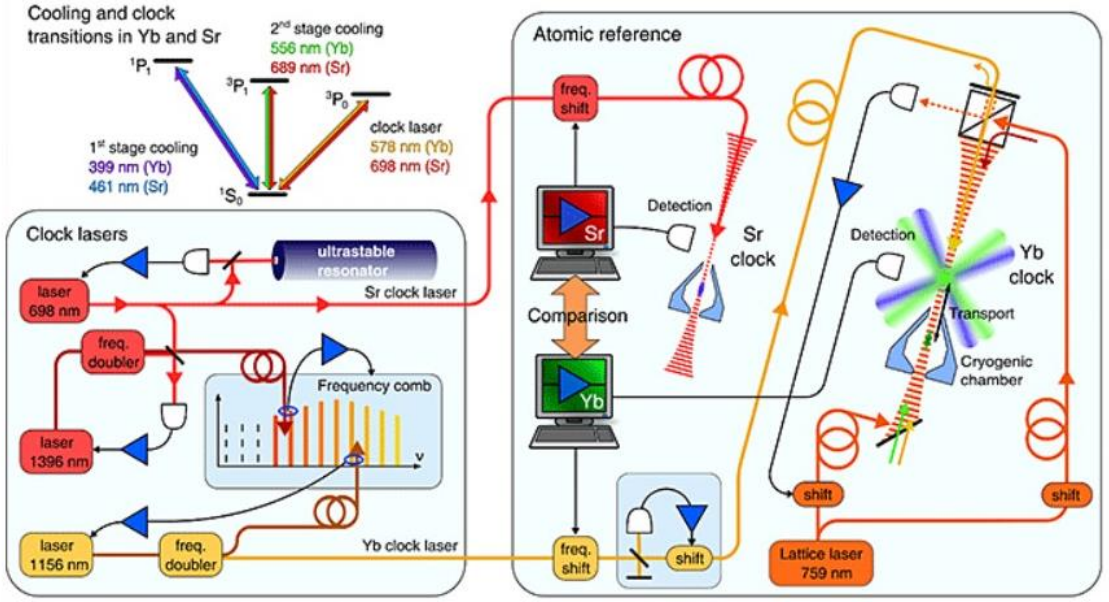


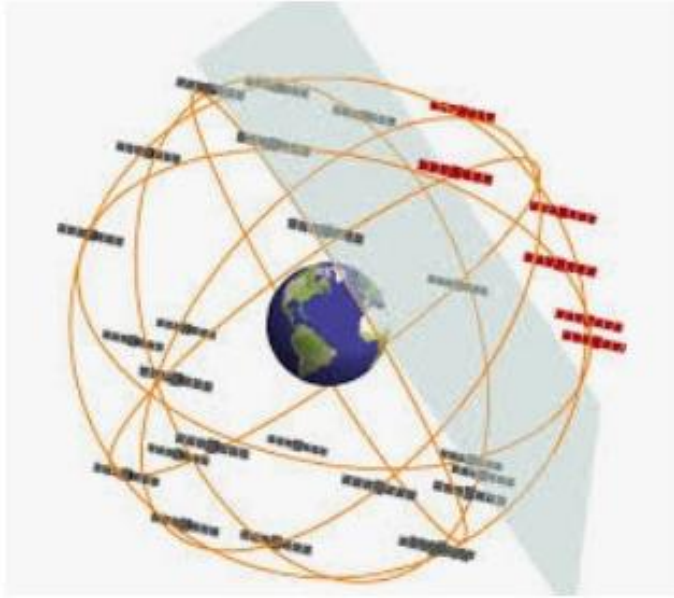
- Fundamental research:
  - Investigating relativity (gravitational redshift, time dilation)
  - Variation of fundamental constants
  - Search for dark matter



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# Variation of fundamental constants

- **Fundamental constant:** any parameter that cannot be calculated with our present knowledge of physics, i.e., a free parameter of a current theory*

- **Examples:** electron charge (***e***), electron mass ($m_e$), proton mass ($m_p$), Plank constant ($\hbar$), velocity of light in vacuum (**c**), Newton constant (***G***), Avogadro constant ($N_A$), Boltzmann constant ($k_B$), and the permeability and permittivity of space ($\varepsilon_0$ and $\mu_0$)

- The number of free parameters depends on the physical model at hand (for example, the minimal standard model of particle physics plus gravitation depends on 20 free parameters)

* J-P.Uzan, Reviews of Modern Physics **75**, 403 (2003)

TABLE I An abbreviated list of the CODATA recommended values of the fundamental constants of physics and chemistry based on the 2014 adjustment.

| Quantity | Symbol | Numerical value | Unit | Relative std. uncert. $u_r$ |
|---|---|---|---|---|
| speed of light in vacuum | $c, c_0$ | 299 792 458 | $\mathrm{m\ s^{-1}}$ | exact |
| magnetic constant | $\mu_0$ | $4\pi \times 10^{-7}$ | $\mathrm{N\ A^{-2}}$ | |
| | | $= 12.566\,370\,614... \times 10^{-7}$ | $\mathrm{N\ A^{-2}}$ | exact |
| electric constant $1/\mu_0 c^2$ | $\epsilon_0$ | $8.854\,187\,817... \times 10^{-12}$ | $\mathrm{F\ m^{-1}}$ | exact |
| Newtonian constant of gravitation | $G$ | $6.674\,08(31) \times 10^{-11}$ | $\mathrm{m^3\ kg^{-1}\ s^{-2}}$ | $4.7 \times 10^{-5}$ |
| Planck constant | $h$ | $6.626\,070\,040(81) \times 10^{-34}$ | J s | $1.2 \times 10^{-8}$ |
| $h/2\pi$ | $\hbar$ | $1.054\,571\,800(13) \times 10^{-34}$ | J s | $1.2 \times 10^{-8}$ |
| elementary charge | $e$ | $1.602\,176\,6208(98) \times 10^{-19}$ | C | $6.1 \times 10^{-9}$ |
| magnetic flux quantum $\hbar/2e$ | $\Phi_0$ | $2.067\,833\,831(13) \times 10^{-15}$ | Wb | $6.1 \times 10^{-9}$ |
| conductance quantum $2e^2/h$ | $G_0$ | $7.748\,091\,7310(18) \times 10^{-5}$ | S | $2.3 \times 10^{-10}$ |
| electron mass | $m_e$ | $9.109\,383\,56(11) \times 10^{-31}$ | kg | $1.2 \times 10^{-8}$ |
| proton mass | $m_p$ | $1.672\,621\,898(21) \times 10^{-27}$ | kg | $1.2 \times 10^{-8}$ |
| proton-electron mass ratio | $m_p/m_e$ | 1836.152 673 89(17) | | $9.5 \times 10^{-11}$ |
| fine-structure constant $e^2/4\pi\epsilon_0\hbar c$ | $\alpha$ | $7.297\,352\,5664(17) \times 10^{-3}$ | | $2.3 \times 10^{-10}$ |
| inverse fine-structure constant | $\alpha^{-1}$ | 137.035 999 139(31) | | $2.3 \times 10^{-10}$ |
| Rydberg constant $\alpha^2 m_e c/2h$ | $R_\infty$ | 10 973 731.568 508(65) | $\mathrm{m^{-1}}$ | $5.9 \times 10^{-12}$ |
| Avogadro constant | $N_A, L$ | $6.022\,140\,857(74) \times 10^{23}$ | $\mathrm{mol^{-1}}$ | $1.2 \times 10^{-8}$ |
| Faraday constant $N_A e$ | $F$ | 96 485.332 89(59) | $\mathrm{C\ mol^{-1}}$ | $6.2 \times 10^{-9}$ |
| molar gas constant | $R$ | 8.314 4598(48) | $\mathrm{J\ mol^{-1}\ K^{-1}}$ | $5.7 \times 10^{-7}$ |
| Boltzmann constant $R/N_A$ | $k$ | $1.380\,648\,52(79) \times 10^{-23}$ | $\mathrm{J\ K^{-1}}$ | $5.7 \times 10^{-7}$ |
| Stefan-Boltzmann constant $(\pi^2/60)k^4/\hbar^3 c^2$ | $\sigma$ | $5.670\,367(13) \times 10^{-8}$ | $\mathrm{W\ m^{-2}\ K^{-4}}$ | $2.3 \times 10^{-6}$ |
| | Non-SI units accepted for use with the SI | | | |
| electron volt $(e/\mathrm{C})$ J | eV | $1.602\,176\,6208(98) \times 10^{-19}$ | J | $6.1 \times 10^{-9}$ |
| (unified) atomic mass unit $\frac{1}{12}m(^{12}\mathrm{C})$ | u | $1.660\,539\,040(20) \times 10^{-27}$ | kg | $1.2 \times 10^{-8}$ |

# Are fundamental constants constant?

Various theories and extensions of the Standard Model predict varying fundamental constants.

- Extra space dimensions are a common feature of theories unifying **gravity** with other interactions (Kaluza-Klein, Superstring and M-theories). Any change in size of these dimensions would manifest itself in the 3D world as variation of fundamental constants

- Fundamental constants depend on scalar fields, which vary in space and time. May be related to “dark energy” and accelerated expansion of the Universe.




- Anthropic principle (Goldilock principle):

"...conditions that are observed in the universe must allow the observer to exist."*

Such "fine tuning" of physical laws, necessary for humans to exist, places constraints on the values of fundamental constants.

Example: low-energy resonance in production of carbon from helium in stars (He+He+He=>C). Slightly different (a few percent) fine structure constant $\alpha \equiv e2/\hbar c$: unstable carbon nuclei→ no life

Variation of coupling constants in space provides a natural explanation of the "fine tuning": we appeared in area of the Universe where values of fundamental constants are suitable for our existence (Goldilocks principle).